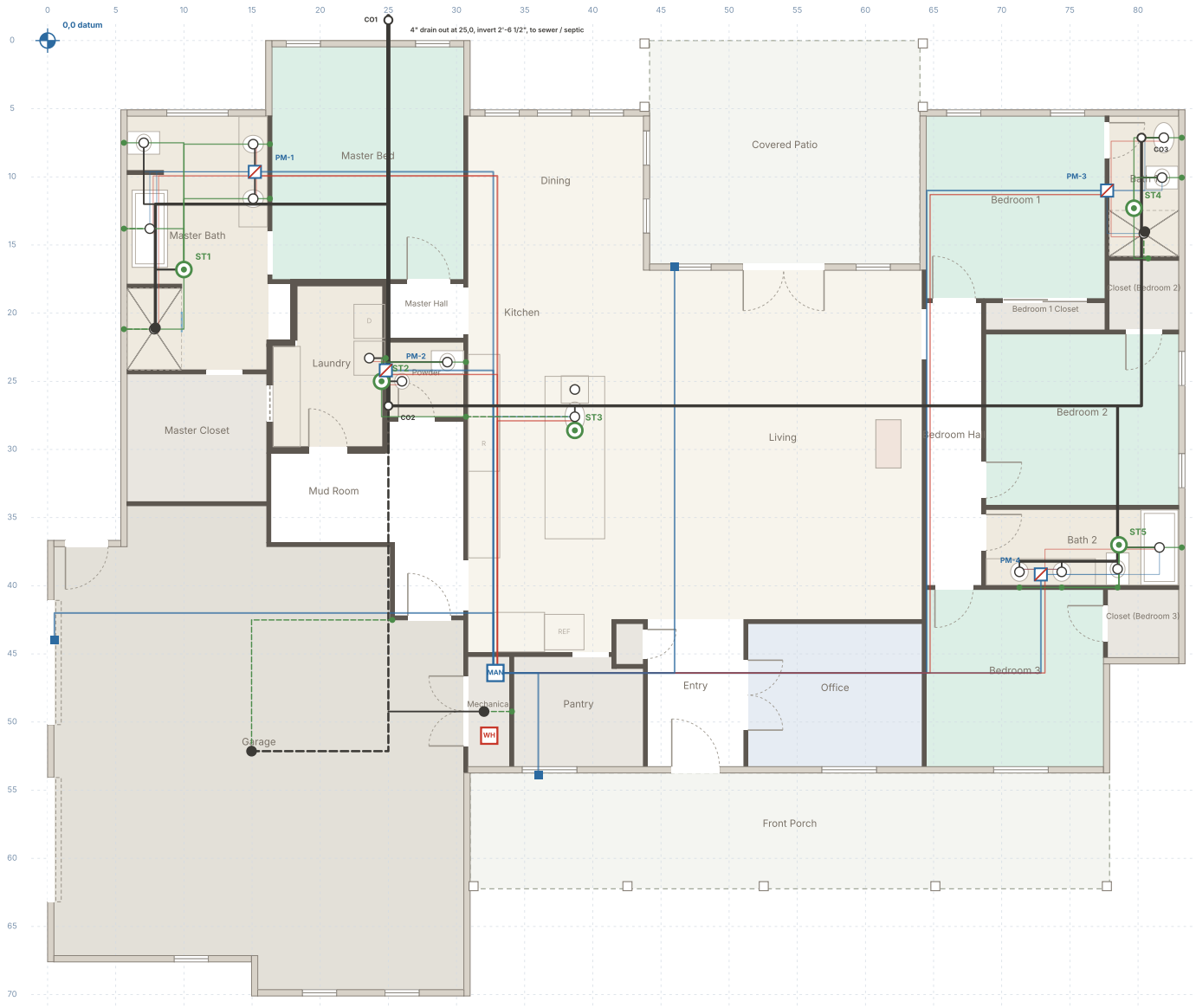
REAR · PATIO



FRONT · PORCH

Drains black, heavier for bigger pipe (4" building drain out the rear at 25,0); floor-drain line dashed; stacks green. · Vents green: a dot where each rises in the wall, thin line across the attic to its stack, dashed where it runs under the slab. · PEX: cold blue, hot red; heavy lines are the group feeds from MAN, light lines the short runs from each group manifold (PM) to its fixtures. · Routing is schematic: the plumber lays out the final runs.

draft

P-1 Plumbing · 64 Stone Sheep Circle

Fixtures from A-5; one drain tree out the rear wall at 25,0, and group PEX feeds. A first draft for the plumber (slab on grade). App version v28. From the owner's plans (Larson House, 6/11/26).

Notes

1. Slab on grade assumed (A-5 shows floor drains in the garage and mechanical closet). All drains and supply runs are under the slab or in walls; no water lines in the attic.
2. One drain tree, every line falling to where the building drain leaves the house at 25,0 (the rear wall line) at an invert of 2'-6 1/2" below the finished floor: a 4" building drain straight out the rear at X 25, a 3" east trunk at Y 26.8 for the kitchen and both east baths, a 3" master bath branch at Y 12, and a 2" line from the garage and mechanical floor drains. 41 DFU in all.
3. Slopes are the IPC minimums (1/8" per ft at 3" and 4", 1/4" per ft below 3"), worked back from the exit; the shallowest invert is 1'-5 1/2" under the slab.
4. Under the slab, top down: 4" of concrete; R-1's 5" nub boards with the radiant loops on them (9" to their underside); R-1's 1" feed pairs right under the boards; then P-1 in the base course, each in its own band so nothing crosses at the same depth: cold PEX about 1'-0" down, hot PEX about 1'-1 1/2" down, and every drain below both (crowns no higher than 1'-2 1/2"). Vent legs under the slab leave their drain at its own depth. Rough in, test and backfill the plumbing before the vapor barrier and boards go down; sleeve each line where it comes up through the boards and slab. R-1's tube runs under the fixtures at 6" centres, so each riser that can move (lav and sink drains, the washer box, the PEX stub-ups and group manifolds, CO3) is set a few inches off its fixture to come up between two tubes; the closet flanges, floor and shower drains, tub drains and stacks can't move, and R-1 marks each one where its installer shifts one run over a channel.
5. Supply: 1" service to a main manifold in the mechanical closet, then one cold and hot pair to each fixture group, in one piece under the slab (no fittings under the slab), to a small manifold in the vanity or wet wall; short 1/2" lines from there to each fixture. 726 ft of PEX in all. 25.3 cold + 10.4 hot WSFU. Hot runs under the slab get R-3 closed-cell insulation (IECC R403.5.3), since they sit below the

Open items

- Confirm the drain leaves at 25,0 (assumed): city sewer or septic, and the invert there. The house needs about 2'-6 1/2" of cover at the wall plus the fall to the main or tank.
- Water source (city or well) and where it enters; the main manifold is drawn in the mechanical closet.
- A demand recirculation pump (button or motion sensor at each bath) on the hot side would cut the wait at the far baths to a few seconds; it needs a 1/2" return or the cold line as the return.
- Water heater fuel and type. A heat-pump water heater needs about 700 cu ft of air or a ducted kit; the 3' x 8' mechanical closet is too small without one. A gas unit needs venting and combustion air.
- Powder room sink is not drawn on A-5 (one is shown here, assumed).
- Garage floor drain: sanitary with an interceptor, or to daylight (lot grading).
- Kitchen island vent method (loop vent vs. AAV) depends on the local code.
- Fixture models: A-5 tags the drains K-9135 and K-9136; tub, toilet, faucet and valve models still to pick.

radiant boards. Pressure-balanced or thermostatic valves at the tubs and showers.

6. Hot water takes longest at master bath (about 47 s at a lavatory); see open items for a recirculation pump.
7. Every trap is vented: up the wall behind the fixture and across the attic to its group's 3" stack, which goes through the roof. Wet walls framed 2×6 (see S-1). ADA toilets as drawn: 17-19" bowl height, grab-bar blocking.
8. Water heater in the mechanical closet with a pan and the T&P and pan drain to the closet floor drain.

Fixture schedule

Tag	Fixture	Room	Trap / drain	DFU	Supply	Fed from	Stack
MB-WC	ADA toilet	Master Bath	3"	3	1/2"	PM-1	ST1
MB-TUB	Pedestal tub (faucet)	Master Bath	1-1/2"	2	1/2"	PM-1	ST1
MB-LAV1	Vanity sink 1	Master Bath	1-1/4"	1	1/2"	PM-1	ST1
MB-LAV2	Vanity sink 2	Master Bath	1-1/4"	1	1/2"	PM-1	ST1
MB-SHR	Shower 4' x 6' (glass door)	Master Bath	2"	2	1/2"	PM-1	ST1
MB-FD	Shower drain K-9136	Master Bath	2"	2	—	—	ST1
B1-LAV	Vanity sink	Bath 1	1-1/4"	1	1/2"	PM-3	ST4
B1-WC	ADA toilet	Bath 1	3"	3	1/2"	PM-3	ST4
B1-SHR	Shower 5' x 3'-4"	Bath 1	2"	2	1/2"	PM-3	ST4
B1-FD	Shower drain K-9135	Bath 1	2"	2	—	—	ST4
PR-WC	ADA toilet	Powder	3"	3	1/2"	PM-2	ST2
PR-LAV	Lavatory (not drawn on A-5; assumed)	Powder	1-1/4"	1	1/2"	PM-2	ST2
LA-WASH	Washer (curved front loading)	Laundry	2" standpipe	3	1/2" box	PM-2	ST2
K-SINK	Kitchen sink (island)	Kitchen	1-1/2"	2	1/2"	MAN	ST3
K-DW	Dishwasher (island)	Kitchen	via sink tailpiece	2	3/8" hot	MAN	ST3
B2-LAV1	Vanity sink 1	Bath 2	1-1/4"	1	1/2"	PM-4	ST5
B2-LAV2	Vanity sink 2	Bath 2	1-1/4"	1	1/2"	PM-4	ST5
B2-WC	ADA toilet	Bath 2	3"	3	1/2"	PM-4	ST5
B2-TUB	Standard tub 1 [66W]	Bath 2	1-1/2"	2	1/2"	PM-4	ST5
ME-FD	Floor drain K-9136	Mechanical	2"	2	—	—	own vent
GA-FD	Floor drain K-9136	Garage	2"	2	—	—	own vent

Tag	Fixture	Room	Trap / drain	DFU	Supply	Fed from	Stack
HB1	Hose bibb, front porch (frost-free)	Outside	—	—	3/4"	MAN	—
HB2	Hose bibb, rear patio (frost-free)	Outside	—	—	3/4"	MAN	—
HB3	Hose bibb, garage door wall (frost-free)	Outside	—	—	3/4"	MAN	—

Drain lines

Every line falls to the exit at 25,0. Inverts are below the finished floor.

Line	What it carries	Size	Length	DFU	Invert at the head	Invert at the end
BD	4" building drain	4"	37 ft	41	2'-3 1/2"	2'-8"
DT	East trunk: kitchen, bath 1, bath 2	3"	75 ft	19	1'-6"	2'-3 1/2"
D2	Bath 2 branch	3"	19 ft	7	1'-6 1/2"	1'-8 1/2"
DM	Master bath branch	3"	26 ft	11	2'-2"	2'-5"
DF	Floor drain line, garage and mechanical	2"	35 ft	4	1'-6 1/2"	2'-3 1/2"

Water supply groups

Each pair runs in one piece under the slab from the main manifold (MAN) in the mechanical closet.

Group	Fixtures	Cold	Hot	Feed length	Hot wait
PM-1	Master bath	3/4"	3/4"	54 ft	~47 s
PM-2	Laundry and powder	3/4"	1/2"	30 ft	~12 s
K-SINK	Kitchen island	1/2"	1/2"	25 ft	~9 s
PM-3	Bath 1	3/4"	1/2"	80 ft	~32 s
PM-4	Bath 2	3/4"	1/2"	47 ft	~21 s

Stacks and vents

Stack	Group	Fixtures	DFU	Vent
ST1	Master bath	6	11	3" VTR
ST2	Laundry and powder	3	7	3" VTR
ST3	Kitchen island (island vent)	2	4	loop vent / AAV
ST4	Bath 1	4	8	3" VTR
ST5	Bath 2	4	7	3" VTR

Bill of materials

Rough-order quantities from the routing above, for pricing. The plumber takes off the final list.

Item	Qty	Notes
4" PVC DWV	42 ft	building drain, including 10 ft outside the rear wall (more to the sewer/septic)
3" PVC DWV	155 ft	trunk, branches, stack bases and toilet arms, + 15 %; plus ~60 ft for the four 3" vent stacks
2" PVC DWV	91 ft	floor drains, washer, + vents
1-1/2" PVC DWV	75 ft	lavs, tubs, kitchen sink, + vents
3/4" PEX	413 ft	group feeds and hose bibbs, + 15 %
1/2" PEX	422 ft	group feeds and runs to the fixtures, + 15 %
Pipe insulation, R-3, closed cell	333 ft	on every hot PEX run under the slab (IECC R403.5.3), + 10 %
Main manifold	1	8 cold + 5 hot outlets, main shutoff
Group manifolds	4	1/2" PEX, 3-5 ports with stops, in a vanity or behind an access panel
Water heater	1	50 gal, pan, expansion tank
Toilets	4	ADA height (A-5)
Tub/shower valves	4	pressure-balanced
Shower drains	2	K-9136 (master), K-9135 (bath 1) per A-5
Floor drains	2	garage and mechanical, K-9136 per A-5
Cleanouts	3	two-way at the exit, floor cleanouts at the junction and the head of the trunk
Hose bibbs	3	frost-free